

Demystifying AWS Backup for Amazon RDS for Oracle

by Tony Mullen and Giorgio Bonzi | on 25 MAR 2024 | in [Advanced \(300\)](#), [AWS Backup](#), [RDS for Oracle](#), [Technical How-to](#) | [Permalink](#) | [Comments](#) | [Share](#)

Customers prefer to host their relational database workloads in a managed service such as [Amazon Relational Database Service \(Amazon RDS\)](#) because of the benefits offered by managed services. Although Amazon RDS offers [automatic backups](#) as a native feature, your data retention policies may require longer retention periods than the retention policy supported by Amazon RDS. That's where [AWS Backup](#) comes to the rescue. AWS Backup provides a centralized and simplified approach to backup management, allowing you to streamline your RDS database backups and gain greater control and flexibility over your data protection strategy.

In this post, we explore the benefits and features of AWS Backup specifically tailored for RDS instances. We explore how it simplifies backup operations, enhances data protection, and enables you to meet your recovery objectives efficiently. Additionally, we provide comprehensive instructions on the configuration, customization, and compliance of your backups.

Solution overview

With Amazon RDS automatic backups, you can specify a retention period of 0–35 days, and enable or disable automatic backups for your RDS instances. The backups are taken daily during a configurable backup window if the retention period is set to greater than or equal to 1, and you cannot adjust the backup frequency or retention settings beyond the available options. AWS Backup provides more flexibility in backup configuration. You can define custom backup schedules, retention periods, and backup lifecycle policies based on your specific requirements. It allows you to create granular backup policies and adjust settings according to your business needs.

In the following sections, we demonstrate how to back up an RDS database using AWS Backup. We move our RDS database from automated Amazon RDS backups to AWS Backup with a daily backup retained for 35 days and a monthly backup retained for 1 year.

Create a backup vault

To create a [backup vault](#), complete the following steps:

1. On the AWS Backup console, choose **Create backup vault**.
2. For **Backup vault name**, enter a unique name for the vault.
3. For **Encryption key**, choose an [AWS Key Management Service](#) (AWS KMS) key for your backup encryption. For this post, we use the default key.
4. Choose **Create backup vault**.

Create backup vault [Info](#)

General

Backup vault name

Backup vault name is case sensitive. Must contain from 2 to 50 alphanumeric or '-' characters.

Encryption key [Info](#)

Description

Default key that protects my Backup data when no other key is defined

Account

This account
([REDACTED])

Key ID

626d5af2-6f6e-
4b88-84f1-4c3fc2317595

Status

✓ Enabled

Backup vault tags - *optional*

Tags specified here help organize and track your Backup vault

No tags.

Add new tag

You can add up to 50 more tags.

Cancel

Create backup vault

You can also create the backup vault programmatically. The following [AWS Command Line Interface](#) (AWS CLI) command creates the backup vault in the same way:

```
aws backup create-backup-vault --backup-vault-name aws-backup-blog-value
```

Configure a backup plan

To configure your backup plan, complete the following steps:

1. On the AWS Backup console, choose **Backup plans** in the navigation pane.
2. Choose **Create backup plan**.
3. Select your backup plan option (you can start with a template, build a new plan, or define a plan using JSON).
For this post, we select **Start with a template**.

4. Choose your template. For this post, we choose **Daily-Monthly-1yr-retention**.

You can amend the entries after the template is loaded.

5. For **Backup plan name**, enter a unique name. For this post, we use `aws-backup-blog-plan`.

Backup plan options [Info](#)

☒ **Start with a template**
Create a Backup plan based on a template provided by AWS Backup.

☐ **Build a new plan**
Configure a new Backup plan from scratch.

☐ **Define a plan using JSON**
Modify the JSON expression of an existing backup plan or create a new expression.

Templates

Choose a template plan with existing rules.

Daily-Monthly-1yr-Retention ▼

Backup plan name

aws-backup-blog-plan

Backup plan name is case sensitive. Must contain from 1 to 50 alphanumeric or '-' '_' characters.

► **Tags added to backup plan - optional**

Backup rules [Info](#)

Add backup rule

Delete

Edit

Backup rules specify the backup schedule, backup window, and lifecycle rules.

	Name	Backup vault
☐	DailyBackups	Default
☐	Monthly	Default

Backup rules [Info](#)

Add backup ruleDeleteEdit

Backup rules specify the backup schedule, backup window, and lifecycle rules.

	Name	Backup vault
☐	DailyBackups	Default
☐	Monthly	Default

▼ Advanced backup settings

Application-consistent backup [Info](#)

Enable application-consistent snapshots for the selected third-party software running on EC2. [Prerequisites required.](#) [🔗](#)

☐ Windows VSS

You can assign resources to this Backup plan after the plan has been created.

You can add more rules to this Backup plan after the plan has been created.

CancelCreate plan

We now amend the backup rules to change the backup vault to the one we created in the previous step.

6. Choose **Add backup rule**.

7. For **Backup vault**, choose the vault you created.

8. Select **Enable continuous backups for point-in-time recovery (PITR)**.

This setting means that the Amazon RDS automated backup feature is managed by the AWS Backup service itself.

9. Make any additional adjustments to your rule and choose **Create Plan**.

Backup plan options

Info

☒ Start with a template

Create a Backup plan based on a template provided by AWS Backup.

☐ Build a new plan

Configure a new Backup plan from scratch.

☐ Define a plan using JSON

Modify the JSON expression of an existing backup plan or create a new expression.

Templates

Choose a template plan with existing rules.

Daily-Monthly-1yr-Retention

Backup plan name

aws-backup-blog-plan

Backup plan name is case sensitive. Must contain from 1 to 50 alphanumeric or '-' characters.

Tags added to backup plan - optional

Backup rules

Info

Add backup rule

Delete

Edit

Backup rules specify the backup schedule, backup window, and lifecycle rules.

	Name	Backup vault
☐	DailyBackups	Default
☐	Monthly	Default

Backup rules [Info](#)[Add backup rule](#)[Delete](#)[Edit](#)

Backup rules specify the backup schedule, backup window, and lifecycle rules.

	Name	Backup vault
☐	DailyBackups	Default
☐	Monthly	Default

▼ Advanced backup settings**Application-consistent backup** [Info](#)

Enable application-consistent snapshots for the selected third-party software running on EC2. [Prerequisites required.](#) [🔗](#)

☐ Windows VSS

[i](#) You can assign resources to this Backup plan after the plan has been created.

[i](#) You can add more rules to this Backup plan after the plan has been created.

[Cancel](#)[Create plan](#)

10. After you have updated the daily rule, choose **Save**.

Edit Backup rule: DailyBackups [Info](#)

Backup rule configuration [Info](#)

Backup rule name

Backup rule name is case sensitive. Must contain from 1 to 50 alphanumeric or '-' characters.

Backup vault [Info](#)

aws-backup-blog-value ▼  [Create new Backup vault](#) 

Backup frequency [Info](#)

Daily ▼

Continuous backups [Info](#)

With continuous backups, you can restore your AWS Backup-supported resource by rewinding it back to a specific time that you choose, within 1 second of precision (going back a maximum of 35 days). Available for RDS, S3, and SAP HANA on Amazon EC2 resources.

☒ Enable continuous backups for point-in-time recovery (PITR)

Backup window

☒ Use backup window defaults - *recommended* [Info](#)
5 AM UTC, starts within 8 hours.

☐ Customize backup window

Transition to cold storage [Info](#)

 Never ▼

Transition to cold is available when the retention period is more than 90 days.

Retention period [Info](#)

Tell AWS Backup how long to store your backups.

35 Days ▼

The retention period for continuous backups can be between 1 and 35 days.

Copy to destination [Info](#)

Choose a Region ▼

► **Tags added to recovery points - optional**

AWS Backup copies tags from the protected resource to the recovery point upon creation. You can specify additional tags to add to the recovery point.

[Cancel](#) [Save](#)

11. Amend the monthly backup to start on day 20 of every month, then choose **Save**.

Edit Backup rule: Monthly [Info](#)

Backup rule configuration [Info](#)

Backup rule name

Backup rule name is case sensitive. Must contain from 1 to 50 alphanumeric or '-' characters.

Backup vault [Info](#)

 [Create new Backup vault](#)

Backup frequency [Info](#)

on

☐ **Enable continuous backups for point-in-time recovery (PITR)** [Info](#)

With continuous backups, you can restore your AWS Backup-supported resource by rewinding it back to a specific time that you choose, within 1 second of precision (going back a maximum of 35 days). Available for RDS, S3, and SAP HANA on Amazon EC2 resources.

Backup window

☒ **Use backup window defaults - recommended** [Info](#)

5 AM UTC, starts within 8 hours.

☐ **Customize backup window**

Transition to cold storage [Info](#)

Retention period [Info](#)

Copy to destination [Info](#)

► **Tags added to recovery points - optional**

AWS Backup copies tags from the protected resource to the recovery point upon creation. You can specify additional tags to add to the recovery point.

[Cancel](#) [Save](#)

Now we must assign resources to the plan. You have the flexibility to add all resources, only specific resources, or even select or exclude resources using tags.

12. For this post, select **Include specific resource types**.

13. For **Resource type**, choose **RDS**.

14. For **Database names**, choose your database (`blog-aws-backup`).

15. Choose **Assign resources**.

Resource selection [Info](#)
Assign resources to this Backup plan using tags and resource IDs.

1. Define resource selection [Info](#)
Protect all resources or specify resources by type or ID.

☐ Include all resource types
Protect all resource types that are enabled in your account.

☒ Include specific resource types
Choose resources by type or specify individual resources by ID.

2. Select specific resource types [Info](#)
Choose specific resource types that you want to protect with this backup plan. You can also exclude specific resource IDs from the selection.

Select resource types ▼

Resource type

Database names

RDS

Choose resources ▼

Remove

blog-aws-backup ✕

3. Exclude specific resource IDs from the selected resource types - optional [Info](#)
Select specific resource IDs that you want to exclude from this assignment.

Select resource types ▼

4. Refine selection using tags - optional [Info](#)
Filter resources by tags. For multiple tags, resources will only be assigned to the backup plan if they satisfy all tag conditions.

Key

Condition for value

Value

🔍 Enter key

Select a condition ▼

🔍 Enter value

Remove

Add tags

You can add up to 29 more tags.

Cancel

Assign resources

After you have configured your backup rules and assigned resources, you can view the plan details on the AWS Backup console.

aws-backup-blog-plan
Delete
View JSON

Summary

Backup plan name aws-backup-blog-plan	Version ID Mzk3ZmNkNWYtYjk1Ni00ZTE0LWlONmQtOTIzNjRhNzI5ZDcw	Last modified July 18, 2023, 14:49:15 (UTC+01:00)	Last runtime -
Backup plan ID 20ce0208-d4ae-4e60-a983-03112da67eb7			

Backup rules (2)

Backup rules specify the backup schedule, backup window, and lifecycle rules.

EditDeleteAdd backup rule

	Name	Backup vault	Destination Backup vault
☐	DailyBackups	aws-backup-blog-value	-
☐	Monthly	aws-backup-blog-value	-

Resource assignments (1)

Resource assignments specify which resources will be backed up by this Backup plan.

DeleteAssign resources

< 1 > ⚙

	Name	IAM role ARN	Creation time
☐	aws-backup-blog-resources	arn:aws:iam::[REDACTED]:role/service-role/AWSBackupDefaultServiceRole	July 18, 2023, 14:53:38 (UTC+01:00)

You can set up the same configuration programmatically, using JSON documents and the AWS CLI. The following JSON document contains the specifics of `aws-backup-blog-plan`:

```
{
  "BackupPlan": {
    "BackupPlanName": "aws-backup-blog-plan",
    "Rules": [
      {
        "RuleName": "DailyBackups",
        "TargetBackupVaultName": "aws-backup-blog-value",
        "ScheduleExpression": "cron(0 5 ? * * *)",
        "StartWindowMinutes": 400,
        "CompletionWindowMinutes": 10080,
        "Lifecycle": {
          "DeleteAfterDays": 35
        },
        "EnableContinuousBackup": true
      },
      {
        "RuleName": "Monthly",
        "TargetBackupVaultName": "aws-backup-blog-value",
        "ScheduleExpression": "cron(0 5 20 * ? *)",

```

Assuming you store the preceding JSON document as `test-backup-plan.json`, you can use the following command to create the backup plan in AWS Backup:

```
$ aws backup create-backup-plan --cli-input-json file:///home/test-backup-plan.json
```

The following JSON document contains the resource assignment specifications to the backup plan `BACKUP-PLAN-ID`; this is the plan ID returned by the previous command:

```
{
  "BackupPlanId": "BACKUP-PLAN-ID",
  "BackupSelection": {
    "SelectionName": "aws-backup-blog-resources",
    "IamRoleArn": "arn:aws:iam::<aws-account-id>:role/service-role/AWSBackupDefault",
    "Resources": [
      "arn:aws:rds:<aws-region>:<aws-account-id>:db:blog-aws-backup"
    ]
  }
}
```

Assuming this JSON is stored as `backup-selection.json`, you can assign these resources to your backup plan using the following command:

```
$ aws backup create-backup-selection --cli-input-json file:///home/backup-selector
```

Monitor backup jobs

To monitor your backup job, complete the following steps:

1. On the AWS Backup console, choose **Jobs** in the navigation pane.
2. You can filter by your specific resource name (in our case, `blog-aws-backup`) to view the successful runs of your job.

AWS Backup > **Jobs**

Jobs

In jobs, you can monitor the status and other details of backup, restore, and copy activity.

Backup jobs | Restore jobs | Copy jobs

Backup jobs Info

Records of your scheduled or on-demand backups.

Filter backup jobs by job ID, status, resource ID or resource type 7 matches

blog-aws-backup X Clear filters

	Backup job ID	Status	Resource name	Resource ID	Resource type	Creation time	Start by
☐	BA5AFD2B-3277-08F1-9332-D64A78CE11CC	Completed	blog-aws-backup	blog-aws-backup	RDS	August 2, 2023, 07:00:00 (UTC+02:00)	August 2, 2023
☐	C58FDF5B-F2B7-4911-293E-54BE5A2CF2E1	Completed	blog-aws-backup	blog-aws-backup	RDS	August 1, 2023, 07:00:00 (UTC+02:00)	August 1, 2023
☐	67E72344-DD7E-C31A-21F1-7899BEBDC85D	Completed	blog-aws-backup	blog-aws-backup	RDS	July 31, 2023, 07:00:00 (UTC+02:00)	July 31, 2023
☐	5286CAF4-F73C-F2C9-55CC-4A0ABCDE0C09	Completed	blog-aws-backup	blog-aws-backup	RDS	July 30, 2023, 07:00:00 (UTC+02:00)	July 30, 2023
☐	AEAC3167-39F9-ADA6-A92C-5CC74C5A2A16	Completed	blog-aws-backup	blog-aws-backup	RDS	July 29, 2023, 07:00:00 (UTC+02:00)	July 29, 2023
☐	00A6C35D-A301-992D-9785-EE4D5244A268	Completed	blog-aws-backup	blog-aws-backup	RDS	July 28, 2023, 07:00:00 (UTC+02:00)	July 28, 2023
☐	9D38FE1F-477B-F9FA-75E2-72C4A459549C	Completed	blog-aws-backup	blog-aws-backup	RDS	July 27, 2023, 07:00:00 (UTC+02:00)	July 27, 2023

3. To view your recovery points, choose **Backup vaults** in the navigation pane and choose the vault you created.

Recovery points (15) Info

Filter by resource type, recovery point ID, status, resource ID or source account ID

Deselect all Actions

	Recovery point ID	Status	Resource name	Resource ID	Resource type	Backup type	Creation time
☐	continuous:db-p7yjm4abilgccposh4pkqk6xse-33801c3a	Available	blog-aws-backup	blog-aws-backup	RDS	Continuous	August 2, 2023, 07:00:00 (UTC+02:00)
☐	awsbackupjob-c58fdf5b-f2b7-4911-293e-54be5a2cf2e1	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	August 1, 2023, 07:00:00 (UTC+02:00)
☐	awsbackupjob-67e72344-dd7e-c31a-21f1-7899bebd85d	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	July 31, 2023, 07:00:00 (UTC+02:00)
☐	awsbackupjob-5286caf4-f73c-f2c9-55cc-4a0abcde0c09	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	July 30, 2023, 07:00:00 (UTC+02:00)
☐	awsbackupjob-aeac3167-39f9-ada6-a92c-5cc74c5a2a16	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	July 29, 2023, 07:00:00 (UTC+02:00)
☐	awsbackupjob-00a6c35d-a301-992d-9785-ee4d5244a268	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	July 28, 2023, 07:00:00 (UTC+02:00)
☐	awsbackupjob-9d38fe1f-477b-f9fa-75e2-72c4a459549c	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	July 27, 2023, 07:00:00 (UTC+02:00)
☐	awsbackupjob-46fb2086-11d6-c5f8-1228-351dcf260041	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	July 26, 2023, 07:00:00 (UTC+02:00)
☐	awsbackupjob-d6110cbd-ef22-78f1-e893-e9554291e138	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	July 25, 2023, 07:00:00 (UTC+02:00)
☐	awsbackupjob-baf707e2-3ef1-d83e-3412-64396cbce5c	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	July 24, 2023, 07:00:00 (UTC+02:00)

Note that the **Recovery point ID** value with the prefix **awsbackup:** and **Backup type** as **Snapshot** is related to the manual snapshot managed by the AWS Backup Monthly backup rule. The ID with the prefix **continuous:** and **Backup type** as **Continuous** is related to the DailyBackups backup rule, which we activated for continuous backups for PITR. As previously mentioned, when you activate this option, AWS Backup takes control of the Amazon RDS automated backup feature.

4. To verify this, on the Amazon RDS console, navigate to the **Maintenance & backups** tab for the specific RDS instance. You will see a message similar to the following screenshot.

Backups [Info](#)

Your automated backups are being managed in a backup plan within AWS Backup. You can still take snapshots manually from this page.


Backup plan exists
 This instance has a backup plan in AWS Backup. To view and manage it, go to the [Backup plans page](#) in AWS Backup.

From a recovery perspective, creating a restore of the instance is straightforward. You have the option to select a continuous restore point ID to perform a PITR or select a snapshot restore point ID to run a snapshot restore.

5. For this post, select the snapshot restore point ID and on the **Actions** menu, choose **Restore**.

Recovery points (1/15) [Info](#)

Filter by resource type, recovery point ID, status, resource ID or source account ID

	Recovery point ID	Status	Resource name	Resource ID	Resource type	Backup type	Creation time
☐	continuous:db-p7ym4abilgccposh4pkqk6xse-33801c3a	Available	blog-aws-backup	blog-aws-backup	RDS	Continuous	August 1, 2023, 07:00:00
☒	awsbackup:job-c58df5b-f2b7-4911-293e-54be5a2cf2e1	Completed	blog-aws-backup	blog-aws-backup	RDS	Snapshot	August 1, 2023, 07:00:00

Actions: Copy, Edit, **Restore**, Delete

- After making your selections, you are presented with options to choose the characteristics of the restored instance.
- After you finalize your choices, initiate the restore process by choosing **Restore backup**.

Protected resource tags [Info](#)

☒ Copy tags from the protected resource to the restored resource

Cancel **Restore backup**

You can observe the restore job being initiated.

AWS Backup > Jobs

Jobs

In jobs, you can monitor the status and other details of backup, restore, and copy activity.

Backup jobs **Restore jobs** Copy jobs

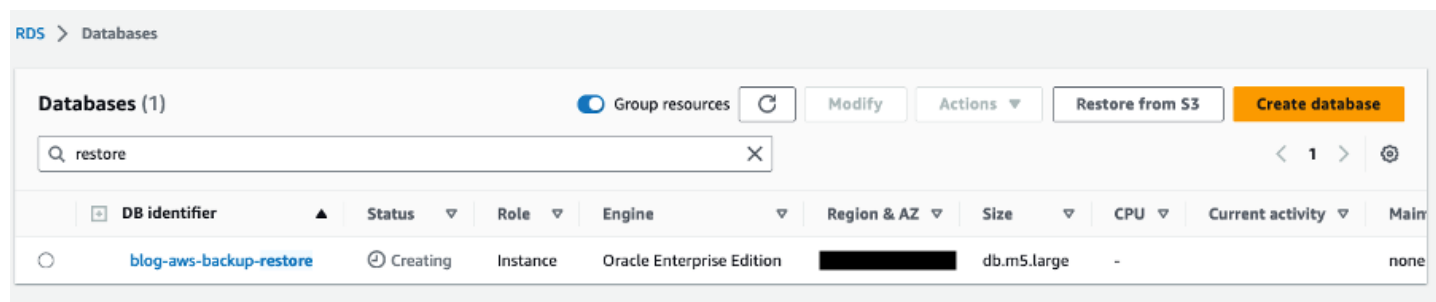
Restore jobs [Info](#)

Records of your backup restoration.

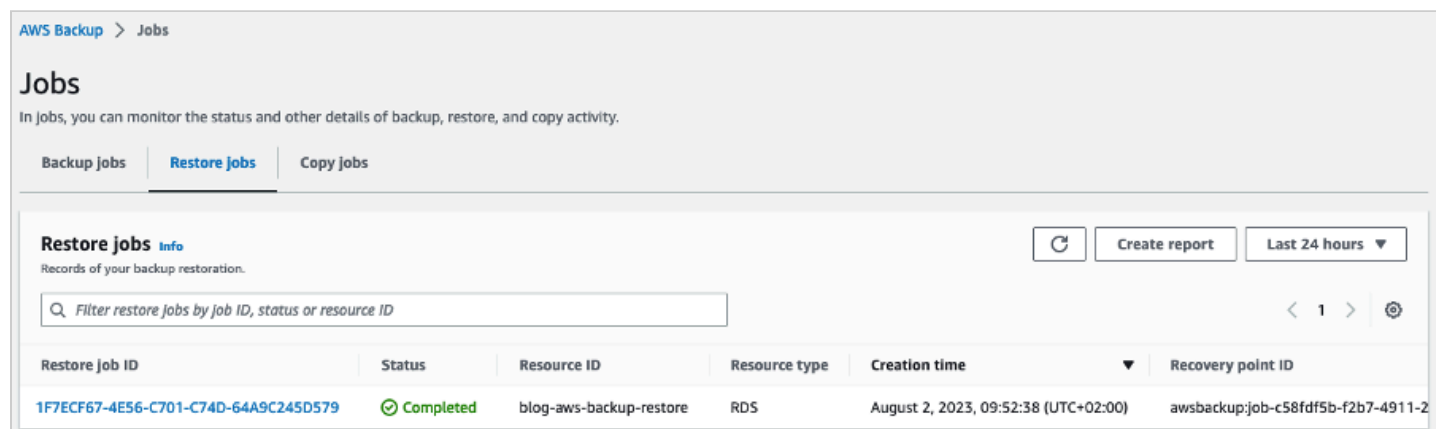
Filter restore jobs by job ID, status or resource ID

Restore job ID	Status	Resource ID	Resource type	Creation time	Recovery point ID
1F7ECF67-4E56-C701-C74D-64A9C245D579	Pending	-	RDS	August 2, 2023, 09:52:38 (UTC+02:00)	awsbackup:job-c58df5b-f2b7-4911-293e-54be5a2cf2e1

You will also see a database created on the Amazon RDS console, which confirms that you restored the recovery point to `blog-aws-backup-restore`.



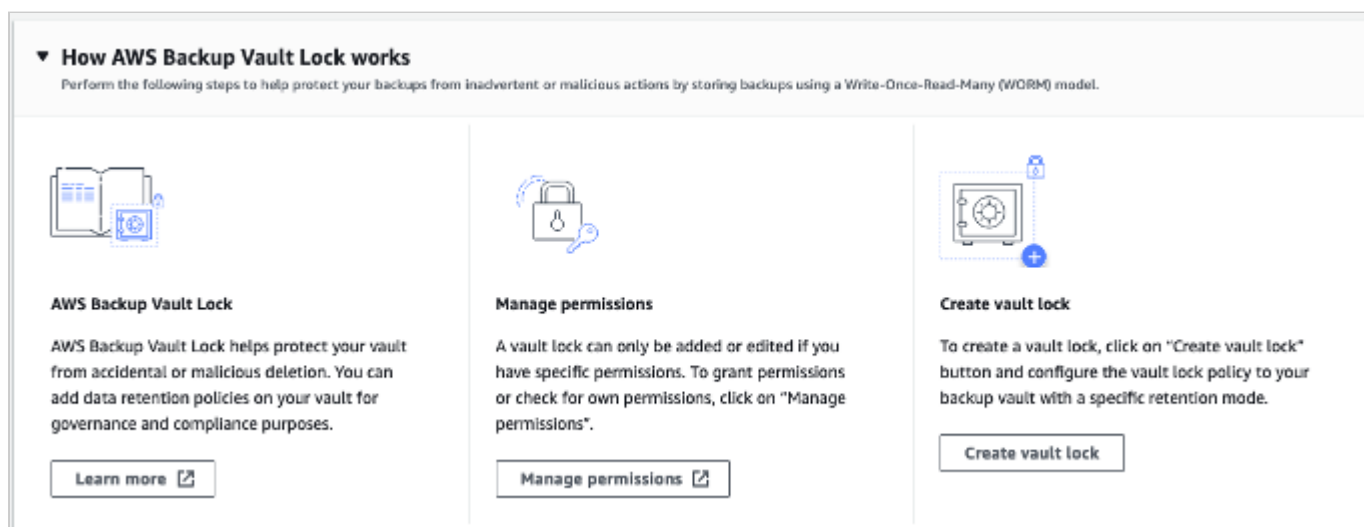
When the restore is complete, the job status changes to **Completed**.



(Optional) Configure a vault lock

Configuring the vault lock is an optional feature that can give you additional security and control over your backup vaults. Each vault can have one vault lock in place.

1. On the AWS Backup console, choose **Create vault lock**.



2. For **Backup vault**, choose the vault you created earlier.
3. Set your vault lock mode:

- a. **Governance mode** – This mode gives access to specific users to manage backup retention. For this post, we select this option.
 - b. **Compliance mode** – When a lock is active in compliance mode and the retention time is over, the vault configuration cannot be altered or deleted by a customer, account or data owner, or AWS.
4. You can optionally configure addition retention periods on top of what you set in the backup plan.

Create vault lock

Vault lock details

Protect your vault with your choice of retention mode.

Backup vault [Info](#)

The vault lock will apply to the contents of the selected vault.

aws-backup-blog-value ▼

Vault lock mode [Info](#)

☒ **Governance mode**

The lock can be managed or deleted by users with specific IAM permissions.

☐ **Compliance mode**

The lock cannot be managed or deleted by any user, even by the root user (account owner) or AWS.

Retention period [Info](#)

Vault locks helps protect backups within the minimum and maximum retention periods.

Minimum retention period - optional [Info](#)

Backups with retention period equal to or greater than the entered value will be protected. 1 day is the default.

1 ▼

Days ▼

Maximum retention period - optional [Info](#)

Backups with a retention period equal to or less than the entered value will be protected.

Enter maximum retention period ▼

Days ▼

i All existing backups in the vault and new backups or copy jobs added to the vault will be protected. This vault can be managed or deleted by only those users with specific IAM permissions. [Learn more](#) [↗](#)

5. Choose **Create vault lock**.

<https://aws.amazon.com/blogs/database/demystifying-aws-backup-for-amazon-rds-for-oracle/>

15/26

6. Enter confirm, select the acknowledgement check box, and choose **Create**.

Create vault lock

**New backups and copy jobs outside of the retention period will fail.**

This vault lock will apply to the backups currently existing in the vault. All new backup or copy jobs within the vault with retention periods shorter than the specified minimum retention period or greater than the maximum retention period will fail. [Learn more](#)

Backups that are within the specified retention period will be protected with [Governance mode](#)

Minimum retention period: **1 Days**



This vault lock can be managed or deleted by only those users with specific IAM permissions. [Learn more](#)

To confirm governance mode, type **confirm** below.

confirm

☐ I acknowledge that the vault lock is configured as desired and will be immutable after the vault is locked

Cancel

Create

You will now see that the vault is locked.

Vault locks (1) [Info](#)

Export as CSV

Manage vault lock

Create vault lock

Search or filter by vault name, lock status, or retention period

Backup vault name	Vault lock status	Minimum retention period	Maximum retention period	Number of backups in the vault
 aws-backup-blog-value	 Locked – Governance mode	1 day	–	1

To configure a vault lock programmatically, use the API `PutBackupVaultLockConfiguration`. The parameters to include will depend on which vault lock mode you want to use. If you create a vault lock in governance mode, do not include `ChangeableForDays`. If this parameter is included, the vault lock will be created in compliance mode.

For this post, we create a vault lock in governance mode with the following code:

```
$ aws backup put-backup-vault-lock-configuration \
  --backup-vault-name aws-backup-blog-value \
  --min-retention-days 7 \
  --max-retention-days 30
```

Copy backups across AWS Regions

With AWS Backup, you can copy backups to multiple Regions either on demand or automatically as part of a scheduled backup plan. This cross-Region replication proves particularly advantageous when adhering to business continuity or compliance requirements necessitating backups to be stored at a minimum distance from production data.

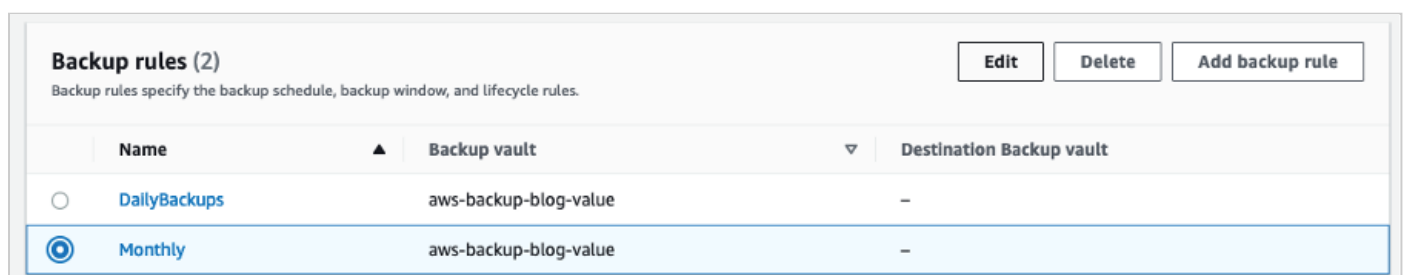
When you copy a backup to a new Region for the first time, AWS Backup performs a full backup replication. Subsequent copies of the same backup within the same target Region become incremental. Furthermore, AWS Backup verifies the re-encryption of your copied data using the customer-managed key of the destination vault.

AWS Backup also allows you to perform backups to multiple AWS accounts, either on demand or through automated scheduling in a backup plan. In a later post, we'll provide a more comprehensive explanation of this feature.

Currently, Amazon RDS [does not support](#) a single copy action that enables both cross-Region and cross-account backup simultaneously. You must choose one option or the other. However, you can use a workaround, described in [Protecting encrypted Amazon RDS instances with cross-account and cross-Region backups](#), using an [AWS Lambda](#) script. The script can listen for the completion of the initial copy, perform the second copy to the desired AWS account or Region, and then delete the first copy.

To schedule a cross-Region backup copy, complete the following steps:

1. Select the backup rule within your backup plan for which you wish to enable the cross-Region backup copy feature, in this case the Monthly rule, then choose **Edit**.



2. For **Copy to destination**, choose the destination Region for your backup copy.
3. For this post, do not enable **Copy to another account's vault**.
4. For **Destination Backup vault**, choose the backup vault located in the destination Region where AWS Backup will store the copied backup.
In this particular scenario, we create a new backup vault to facilitate cross-Region copy.
5. Choose **Create new Backup vault**.

The screenshot shows the 'Copy to destination' configuration page in the AWS Backup console. At the top, there's a section titled 'Copy to destination' with an 'Info' link. Below it is a dropdown menu showing 'Europe (Ireland)' and a 'Remove' button. A toggle switch for 'Copy to another account's vault' is turned off. The 'Destination Backup vault' section has a dropdown showing 'Default', a refresh icon, and a 'Create new Backup vault' button with an external link icon. Below this is the 'Transition to cold storage' section with an 'Info' link, a checkbox, and a dropdown set to 'Never'. The 'Retention period' section has an 'Info' link, a text input with '5', a dropdown set to 'Weeks', and an 'Add copy' button. At the bottom, there's a section titled 'Tags added to recovery points - optional' with a description: 'AWS Backup copies tags from the protected resource to the recovery point upon creation. You can specify additional tags to add to the recovery point.' At the very bottom right are 'Cancel' and 'Save' buttons.

6. For **Backup vault name**, enter a name (for this post, `aws-backup-blog-xvault`).
7. For **Encryption key**, enter your customer-managed key (`aws-backup-blog-key`).

8. Choose **Create backup vault**.

Create backup vault [Info](#)

General

Backup vault name

Backup vault name is case sensitive. Must contain from 2 to 50 alphanumeric or '-' characters.

Encryption key [Info](#)

Description	Account	Key ID	Status
Customer Managed Key used for AWS backup cross-region copy	This account	44226ef6-ad18-4fc9-ac88-8bec9bfdbf19	Enabled

Backup vault tags - optional
Tags specified here help organize and track your Backup vault

No tags associated with this vault.

[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Create backup vault](#)

9. Choose **Save**.

Copy to destination [Info](#)

Europe (Ireland) ▼ Remove

☐ Copy to another account's vault

Destination Backup vault

The vault to which your backup copy will be made.

aws-backup-blog-xvault ▼ ↺ Create new Backup vault ↗

Transition to cold storage [Info](#)

Never ▼

Retention period [Info](#)

Tell AWS Backup how long to store your backups.

5

Weeks ▼

Add copy

► **Tags added to recovery points - *optional***

AWS Backup copies tags from the protected resource to the recovery point upon creation. You can specify additional tags to add to the recovery point.

Cancel Save

10. Check the copy configuration on the Monthly backup rule detail page.

AWS Backup > Backup plans > aws-backup-blog-plan > Monthly

Monthly

DeleteEdit

Summary

Backup rule name

Monthly

Frequency

Monthly

At 09:00 AM UTC, on day 2 of the month

Start within

1 hour

Complete within

2 hours

Lifecycle

Transition to cold storage after 1 month

Expire after 1 year

Backup vault

aws-backup-blog-value

Tags added to recovery points - optional

-

Copy configuration

Destination region	Destination Backup vault	Destination account ID	Transition to cold storage	Retention period
Europe (Ireland)	aws-backup-blog-xvault	-	Never transition to cold storage	Expire after 5 weeks

The following screenshot shows the backup vault detail page.

AWS Backup > Backup vaults > aws-backup-blog-xvault

aws-backup-blog-xvault [Info](#)

[Create vault lock](#)
[Delete vault](#)
[Edit access policy](#)

Summary

Vault name aws-backup-blog-xvault	KMS encryption key ID 44226ef6-ad18-4fc9-ac88-8bec9bfdbf19	Vault lock -
Vault ARN arn:aws:backup:eu-west-1:123456789012:backup-vault:aws-backup-blog-xvault	Creation date August 2, 2023, 10:51:30 (UTC+02:00)	Retention period for vault lock Minimum retention period: - Maximum retention period: -

Recovery points (1) [Info](#)

[Refresh](#)
[Deselect all](#)
[Actions](#)

Filter by resource type, recovery point ID, status, resource ID or source account ID

☐	Recovery point ID	Status	Resource name	Resource ID	Resource type
☐	awsbackup:copyjob-bd4ff1b4-4650-49d2-b6df-bf4fe04f1b58	Completed	blog-aws-backup	blog-aws-backup	RDS

You can set up the same configuration programmatically using JSON documents and the AWS CLI. The following JSON document contains the specifics to update `aws-backup-blog-plan`:

```
{
  "BackupPlan": {
    "BackupPlanName": "aws-backup-blog-plan",
    "Rules": [
      {
        "RuleName": "DailyBackups",
        "TargetBackupVaultName": "aws-backup-blog-value",
        "ScheduleExpression": "cron(0 5 ? * * *)",
        "StartWindowMinutes": 400,
        "CompletionWindowMinutes": 10080,
        "Lifecycle": {
          "DeleteAfterDays": 35
        },
        "EnableContinuousBackup": true
      },
      {
        "RuleName": "Monthly",
        "TargetBackupVaultName": "aws-backup-blog-value",
```

Assuming you store the preceding JSON document as `test-backup-plan.json`, you can run the following command to update the backup plan:

```
$ aws backup update-backup-plan --backup-plan-id "BACKUP-PLAN-ID" --cli-input-json f
```

Clean up

The retention period defined in the AWS Backup rules cleans up expired RDS instance backups. For a complete cleanup of the environment configured for this post, complete the following steps:

1. [Delete the DB instance](#) `blog-aws-backup-restore`, which you restored from a recovery point.
2. [Remove the vault lock](#) `aws-backup-blog-value`.
3. Delete the resource assignment `aws-backup-blog-resources` from `aws-backup-blog-plan`.
4. [Delete the backup plan](#) `aws-backup-blog-plan`.
5. [Delete all the backups](#) in the `aws-backup-blog-value` backup vault.
6. [Delete the backup vault](#) `aws-backup-blog-value`.
7. Delete the `aws-backup-blog-xvault` backup vault in the secondary Region that you configured for the cross-Region copy.

Amazon RDS backup and AWS Backup

Let's run through the primary commonalities and distinctions between the backup options available inside Amazon RDS itself and the functionalities provided by AWS Backup. This post centers its attention primarily on [Amazon RDS for Oracle](#). It's worth noting that these principles can be seamlessly extended to the other available engines: [Amazon RDS for MySQL](#), [Amazon RDS for MariaDB](#), [Amazon RDS for PostgreSQL](#), and [Amazon RDS for SQL Server](#).

Data protection

Amazon RDS does encrypt your RDS DB instances. Data that is encrypted at rest includes the underlying storage for DB instances, its automated backups, read replicas, and snapshots. Amazon RDS uses a KMS key to encrypt these resources. You can use an AWS managed key, or you can create customer managed keys. To manage the customer managed keys used for encrypting and decrypting your Amazon RDS resources, you use AWS KMS.

Within AWS Backup, a [backup vault](#) is a container that stores and organizes your backups. When establishing a backup vault, it's necessary to designate the KMS encryption key that encrypts some of the backups placed in this vault ([independent AWS Backup encryption](#)). For Amazon RDS, independent AWS Backup encryption is not supported. Instead, RDS snapshots are automatically encrypted with the same encryption key that was used to encrypt the source RDS database. Consequently, snapshots of unencrypted RDS databases remain unencrypted. This principle extends to Aurora cluster snapshots. Be aware that AWS managed keys are not supported for AWS Backup [cross-account backup](#). For further insights into this particular subject, refer to [How encryption works in AWS Backup](#).

AWS Backup introduces a feature, [AWS Backup Vault Lock](#), which can be helpful in giving you additional security and control over your backup, bolstering data integrity by helping prevent accidental or unauthorized deletions of backups. When it comes to separation of duties, both services enable the allocation of [roles and permissions](#), but the centralized nature of AWS Backup streamlines this process, promoting clearer access control. In essence, although both Amazon RDS and AWS Backup uphold high standards of data security, the latter introduces enhancements like vault locks and centralized role management, further strengthening data protection measures.

Cost

The cost distinction between the backup capabilities offered by Amazon RDS and those facilitated through AWS Backup revolves around their operational models. Amazon RDS [provides automated backups](#) as an integral part of its service, incurring no additional charges for the feature itself. The cost of a retained automated backup is the cost of the total storage of the system snapshots that are associated with it. The same applies to the [manual snapshots](#) that are always incremental. Conversely, AWS Backup offers a centralized management solution that covers multiple AWS services, including Amazon RDS. Regarding AWS Backup and Amazon RDS, the charges are the same as what you would incur if you were using automated backups for Amazon RDS.

Whether you're using Amazon RDS or AWS Backup, you won't incur backup storage costs as long as the cumulative backup storage remains below 100% of your total database storage size within a given Region. Moreover, there are no supplementary fees for transaction log backups. For further insights into the calculation of Amazon RDS backup costs, see [Demystifying Amazon RDS backup storage costs](#).

When it comes to Amazon RDS [cross-Region automated backups](#) and [DB snapshot copy](#), as well as [AWS Backup cross-Region backups](#), data transfer costs only arise when moving data out of a Region, with no fees for transfers within the same Region. Additionally, charges are applied for the storage of backups copied in the destination Region.

Features

Let's dig deeper into a comparison between the primary backup features offered by Amazon RDS and AWS Backup for Amazon RDS:

- **Continuous backups and PITR transaction log backups** – Both AWS Backup and Amazon RDS use the same mechanism for running [continuous database backups and point-in-time-recovery \(PITR\)](#), with a shared maximum retention period of 35 days. If you use AWS Backup for both Amazon RDS snapshots and continuous backups, AWS Backup will intelligently schedule your backup windows, along with the Amazon RDS maintenance window, to help prevent conflicts. You no longer have to manually schedule one backup window hours before another. You can't control the Amazon RDS automated backup window.
- **Database snapshots** – Amazon RDS requires manual snapshots or custom automation for [database snapshots](#) taken outside the automated backup PITR. In contrast, AWS Backup rules enable you to schedule database snapshots with customizable frequencies, such as daily, weekly, or monthly.
- **Database snapshot retention** – Amazon RDS necessitates manual deletion of outdated database snapshots, whereas AWS Backup offers a more agile approach by allowing you to configure a retention period in addition to the snapshot frequency.

- **Cross-Region backup copies** – AWS Backup permits configuration of cross-Region backup copies for both continuous PITR backups and database snapshots. When a cross-Region copy of a continuous backup is made, the copied recovery point (backup) becomes a snapshot (periodic) backup. PITR is not available for these copies. For Amazon RDS, [automated backup copies can be automated across Regions](#), whereas [manual snapshot copies](#) require extra configurations.
- **Cross-account backup copies** – AWS Backup enables setup of cross-account backup copies for both continuous PITR backups and database snapshots. When a cross-account copy of a continuous backup is made, the copied recovery point (backup) becomes a snapshot (periodic) backup. Amazon RDS requires manual implementation to automate the cross-account backup copy.
- **Monitoring** – Both Amazon RDS and AWS Backup send events to [Amazon CloudWatch Events](#) and [Amazon EventBridge](#), which facilitate monitoring through [Amazon Simple Notification Service](#) (Amazon SNS) subscriptions. AWS Backup further offers [cross-account monitoring capabilities](#) for policies and associated jobs in the management account, enhancing monitoring efficiency.
- **Cost monitoring** – With [cost allocation tags](#), you can effectively track and optimize backup costs for both AWS Backup and Amazon RDS.

Conclusion

In this post, we provided a comprehensive comparison highlighting the capabilities of Amazon RDS and AWS Backup in terms of data protection, cost, and backup features. We showed how to switch your automated Amazon RDS backups to use AWS Backup with extended retention periods. We also showed how to stop those backups from being removed by using a vault lock and how to implement backup copies across Regions.

By using AWS Backup for your Amazon RDS backups, you can simplify your backup operations, achieve greater control over backup configurations, and enhance your data protection and disaster recovery capabilities. Join us in future posts as we explore the various features and benefits of AWS Backup, empowering you to safeguard your RDS databases effectively.

For more information about AWS resources and third-party applications supported by AWS Backup and the features available, see [Feature available by resource](#).

About the authors



Tony Mullen is a Senior Database Specialist Solutions Architect based in Manchester. With a focus on relational database engines, he assists customers in migrating and modernizing their database workloads to AWS.

Giorgio Bonzi is a Senior Database Specialist Solution Architect based in Milan. With a focus on the Oracle database engine, he provides guidance and technical assistance to customers to help them design, deploy, and optimize relational database workloads on AWS.



Comments

1 Comment

 Login ▼

G

Join the discussion...

LOG IN WITH

OR SIGN UP WITH DISQUS 

Name



Share

Best Newest Oldest



Iván García

2 months ago

Great info, thanks!

o

o

Reply



Subscribe

Privacy

Do Not Sell My Data